



University of Alkafeel, College of Dentistry, Year 3, Course 1

Pathology Lab 8 Medical Genetics

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Learning Objectives

- ☐ Know what Medical Genetics is.
- ☐ Know what Chromosomes are and their structure
- ☐ Know what the types of chromosome are.
- ☐ Know how to analyze chromosomes
- ☐ Understand what the abnormalities that occur to chromosomes and their types.

Introduction

Medical genetics is the branch of medicine that studies the role of genetic variation in human disease.

In pathology, it explains the etiology, mechanisms, and morphological consequences of genetic abnormalities at both the molecular and cellular levels.

Chromosome

Chromosomes are long, organized DNA-protein structures found in the nucleus.

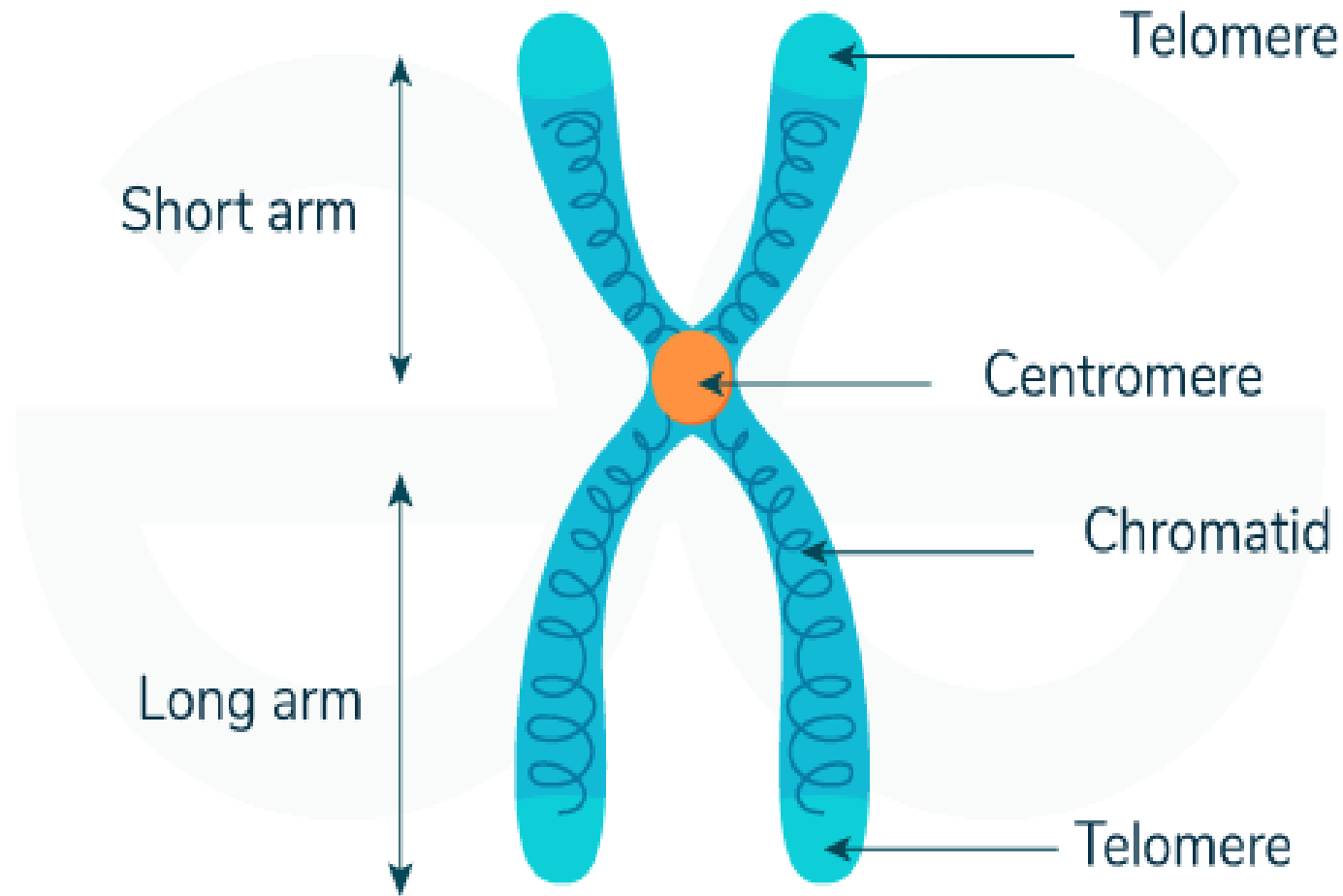
Human Chromosome Number

- 46 chromosomes (23 pairs)
 - 22 pairs of autosomes
 - 1 pair of sex chromosomes (XX or XY)

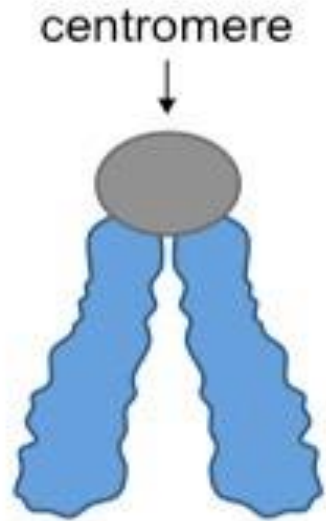
Structure of a Chromosome

- **Chromatid**: one half of a duplicated chromosome
- **Centromere**: constricted region dividing the chromosome into:
 - p arm (short arm)
 - q arm (long arm)
- **Telomeres**: ends of chromosomes, protect from degradation

Chromosome

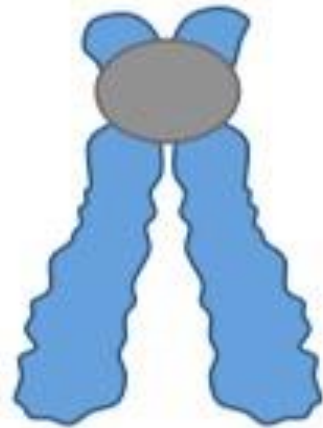


Types of Chromosomes



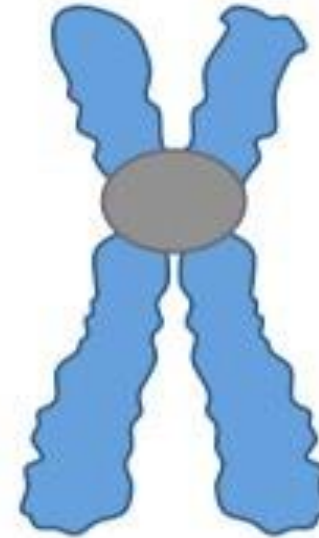
Telocentric
(not in humans)

- Centromere is very **close to the end**
 - Very short p arm, long q arm
- chromosomes:** 13, 14, 15, 21, 22

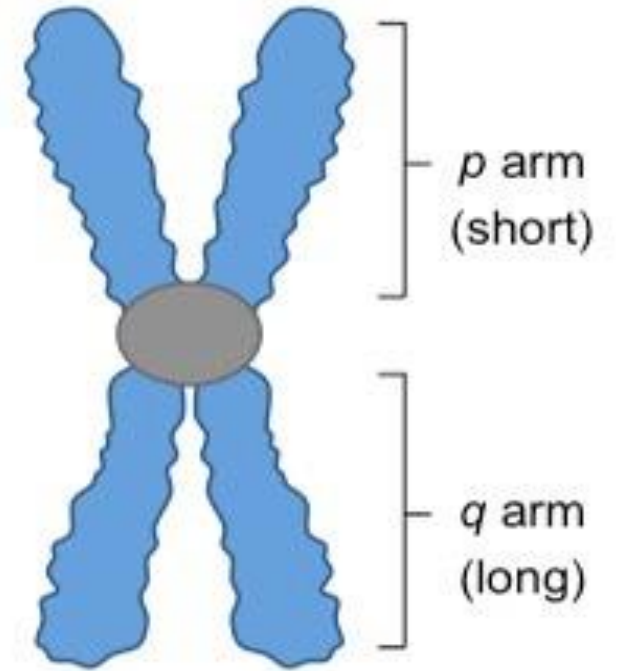


Acrocentric

- Centromere is **slightly off-center**
 - p arm is shorter than q arm
- Examples:** Chromosomes 2, 4–12, 17, 18



Submetacentric



Metacentric

- Centromere is in the **middle**
 - p arm $\approx$ q arm (almost equal length)
- Examples:** Chromosomes 1, 3, 16, 19, 20

Analyzing Chromosomes

Karyotyping is a laboratory method used to look at a person's chromosomes to detect abnormalities.

Step 1: Collect a Sample

The most common sample is **peripheral blood**.

Other possible samples:

- amniotic fluid (prenatal)
- bone marrow (hematologic disorders)



Analyzing Chromosomes

Step 2: Grow the Cells (Culture)

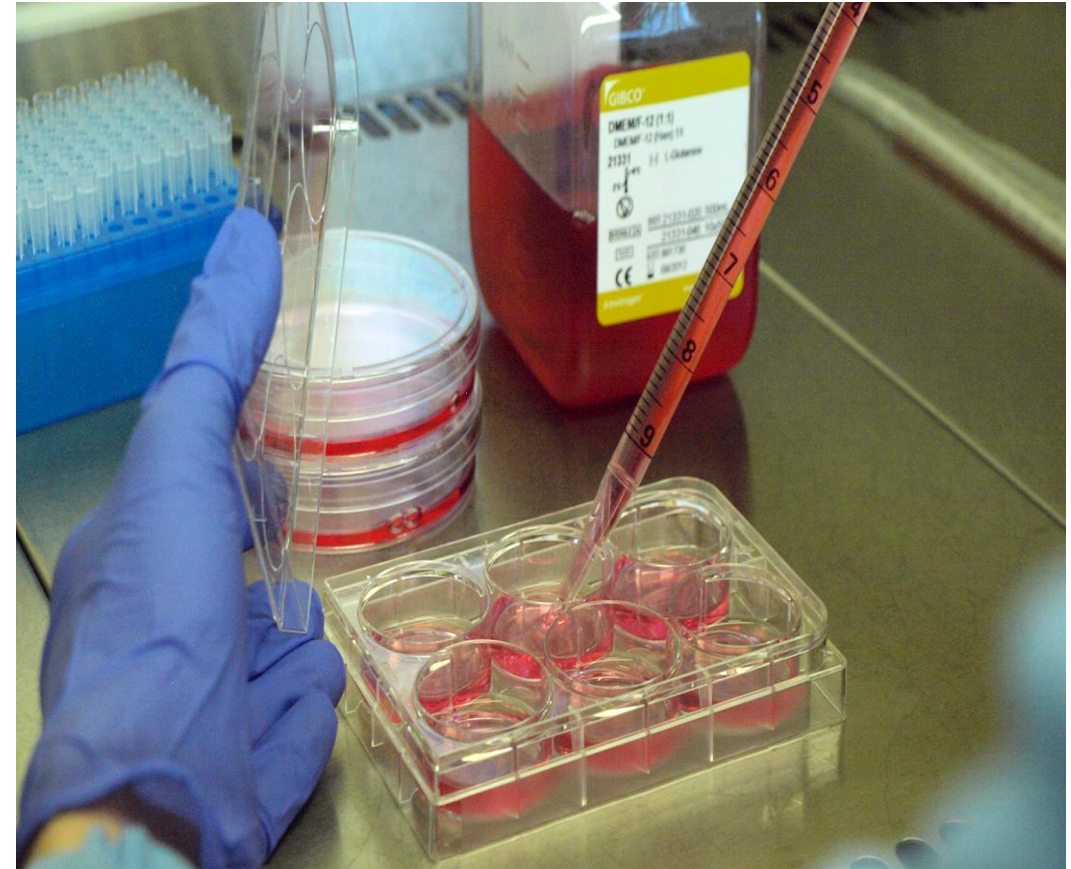
Cells must be **dividing** to see chromosomes.

So the sample is placed in a culture medium for **48–72 hours** to stimulate cell growth.

Step 3: Stop Cells in Metaphase

A chemical called **colchicine** is added.

- It stops cells in **metaphase**, when chromosomes are condensed and visible.

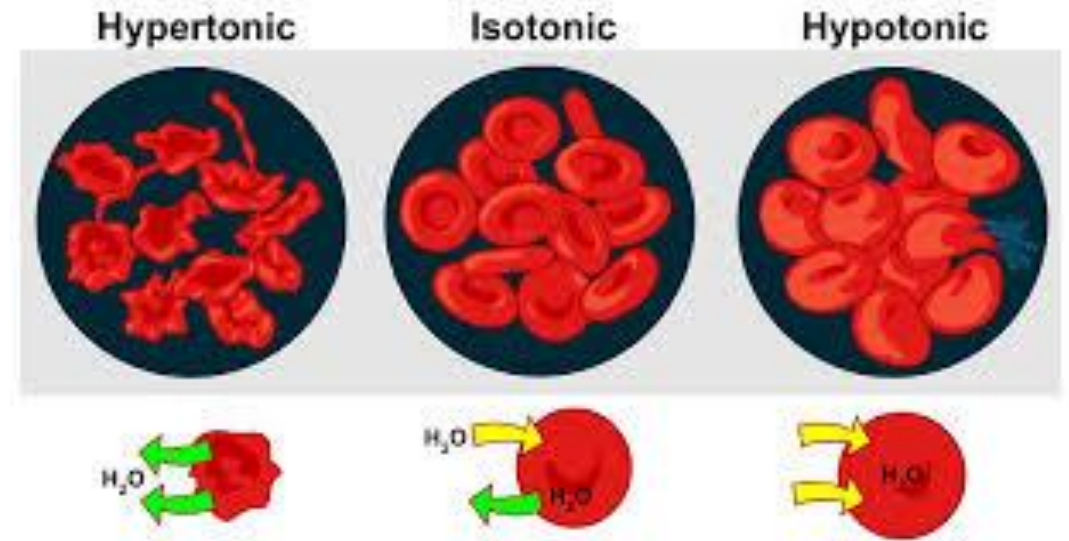


Analyzing Chromosomes

Step 4: Lyse the Cells (Hypotonic Treatment)

Cells are placed in a **hypotonic solution**.

- Water enters the cells
- They swell
- Chromosomes spread out for easier visualization



Step 5: Fix the Cells

A fixative (usually methanol + acetic acid) preserves the cell structure.



Analyzing Chromosomes

Step 6: Drop Cells on a Slide

The cell suspension is dropped onto a glass slide.

When the drop bursts:

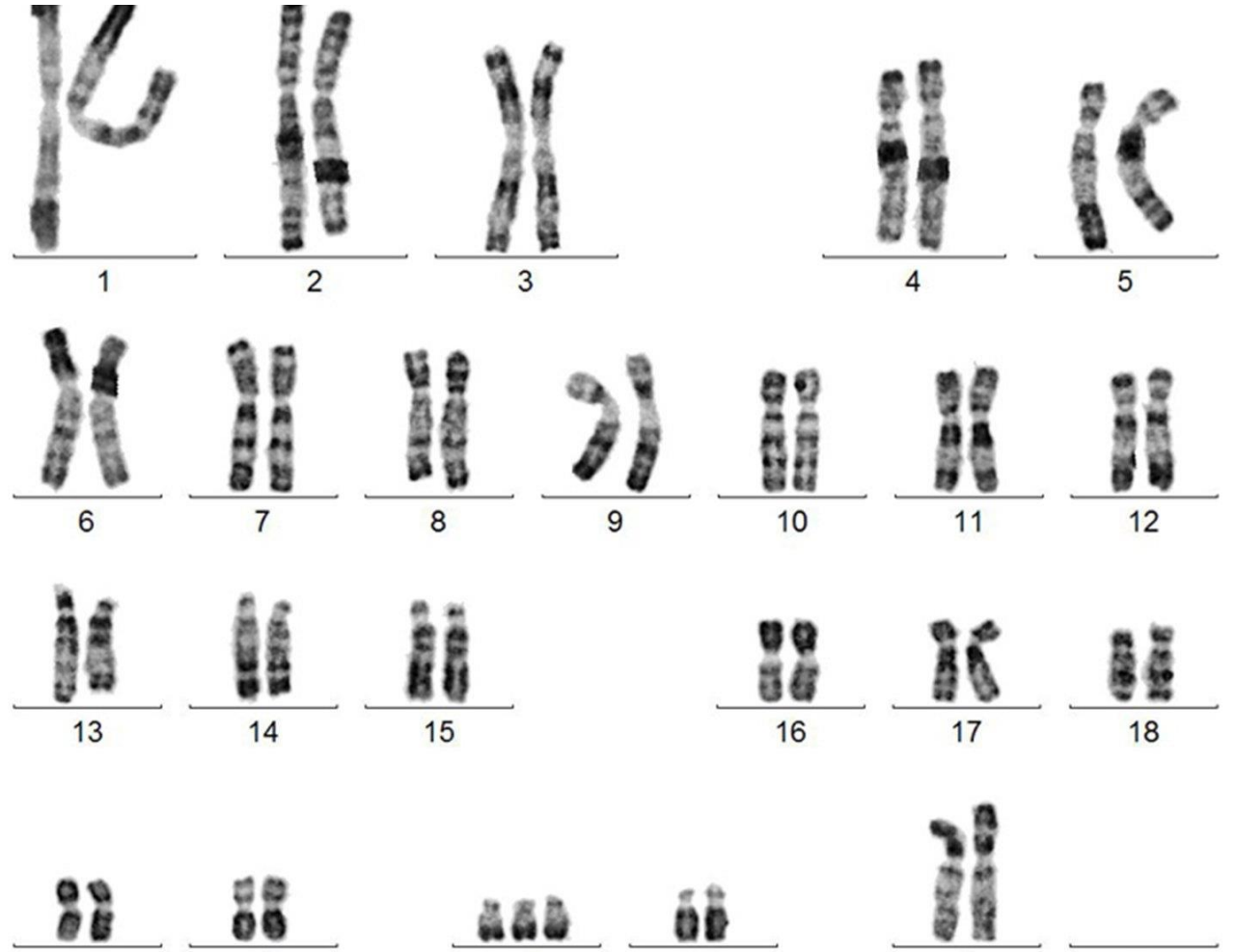
- chromosomes spread apart
- making them easy to analyze

Step 7: Stain the Chromosomes

The slide is stained using **Giemsa stain** → *G-banding*.

This creates a **characteristic pattern of dark and light bands** on each chromosome.

Analyzing Chromosomes





1



2



3



4



5



6



7



8



9



10



11



12



13



14



15



16



17



18



19



20



21



22



X



Y

or



X

X

Shown here are human chromosomes

Chromosomal Abnormalities

Chromosomal abnormalities fall into **two major types**:

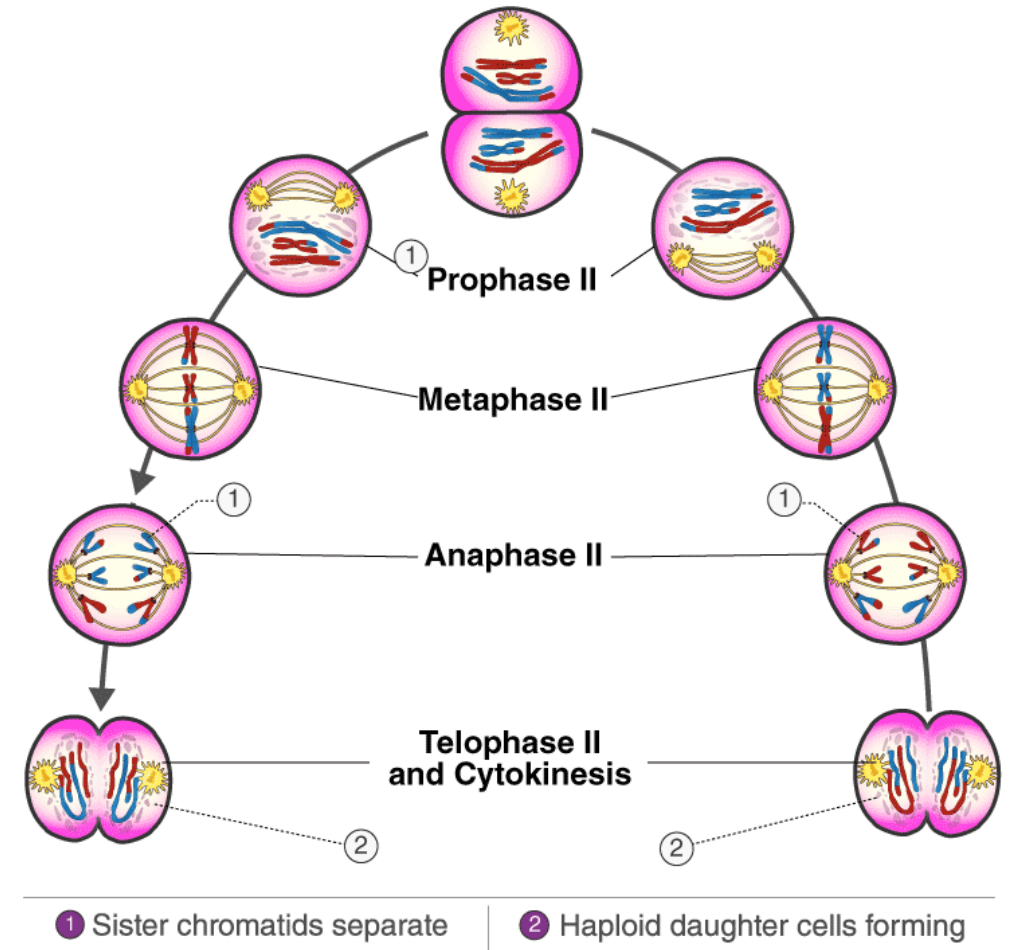
1.Numerical abnormalities (change in chromosome number)

2.Structural abnormalities (change in chromosome structure)

Numerical Abnormalities (Aneuploidy)

Caused by nondisjunction(is the failure of chromosomes to separate properly during **cell division** (meiosis or mitosis)).

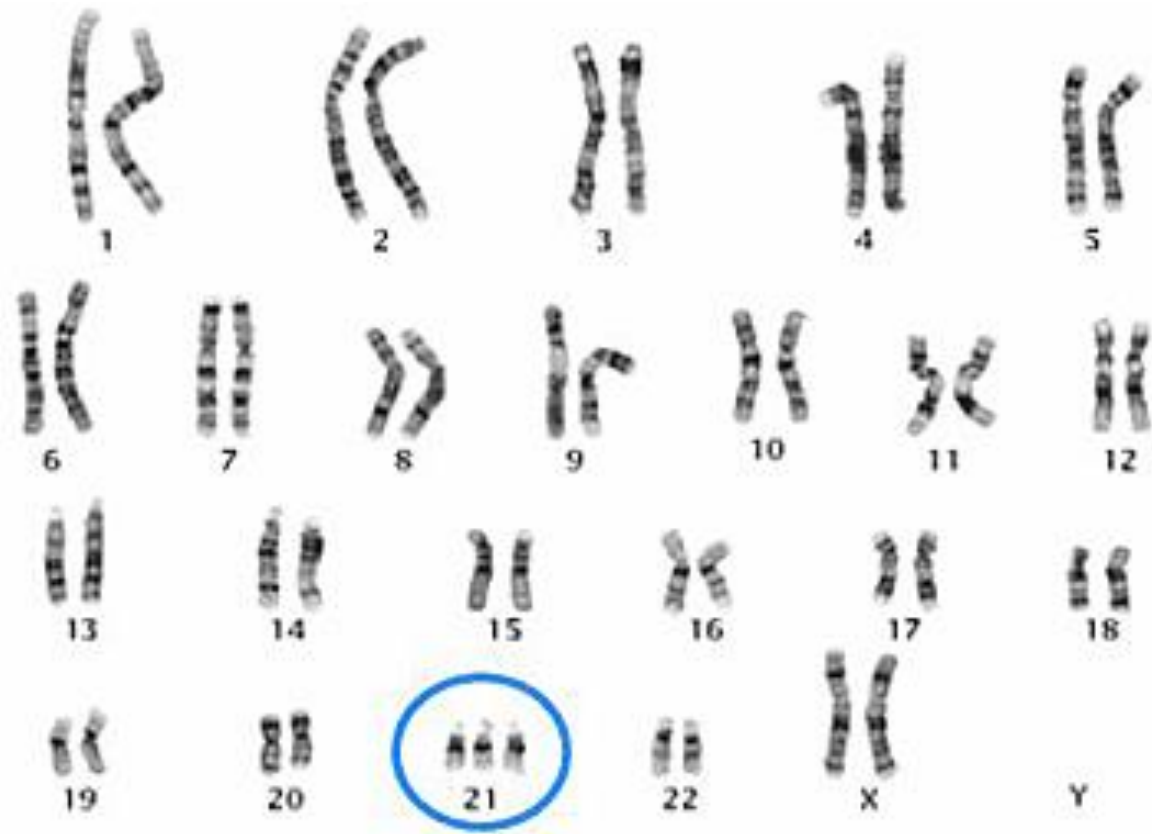
As a result, one daughter cell receives **too many chromosomes**, and the other receives **too few**)



Numerical Abnormalities (Aneuploidy)

1-Autosomal Trisomies

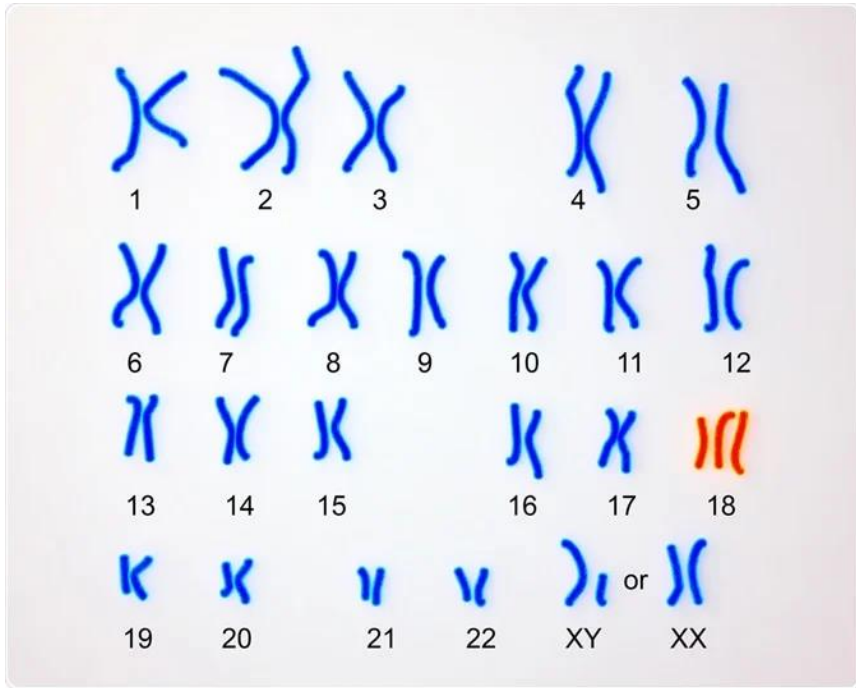
- Trisomy 21 (Down syndrome)



Numerical Abnormalities (Aneuploidy)

1-Autosomal Trisomies

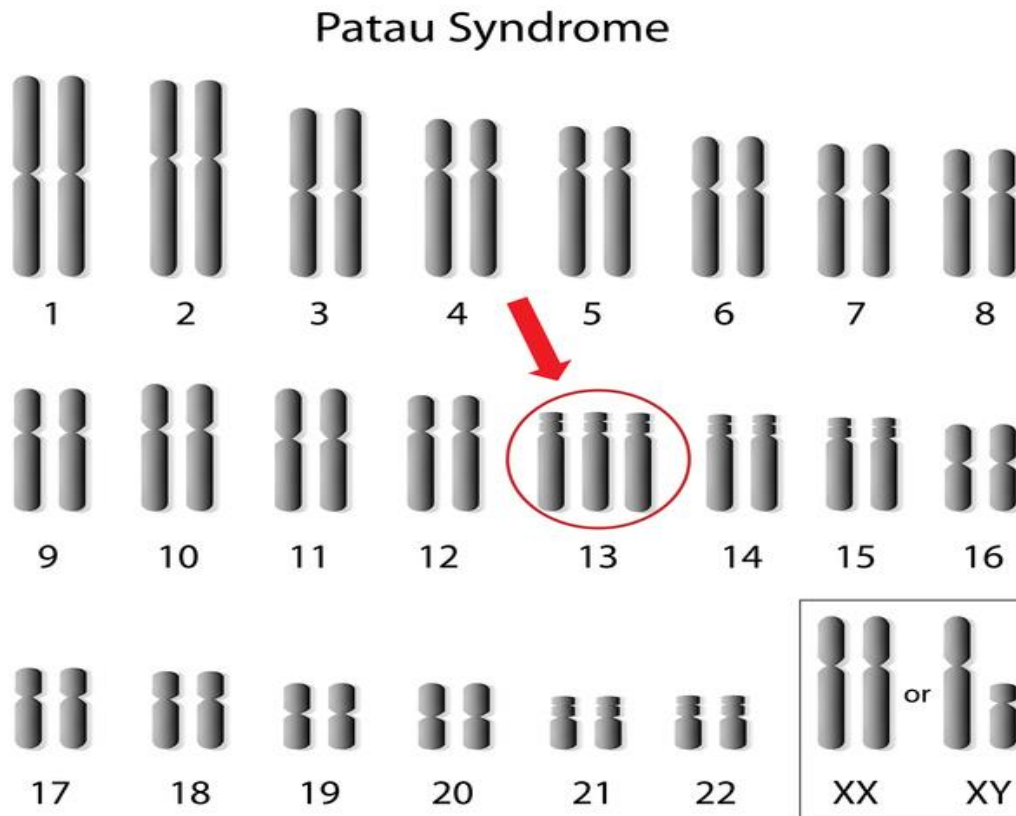
•Trisomy 18 (Edwards syndrome)



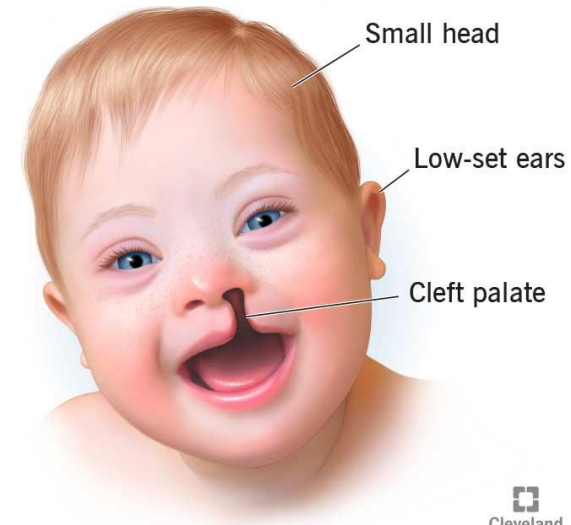
Numerical Abnormalities (Aneuploidy)

1-Autosomal Trisomies

•Trisomy 13 (Patau syndrome)



Trisomy 13

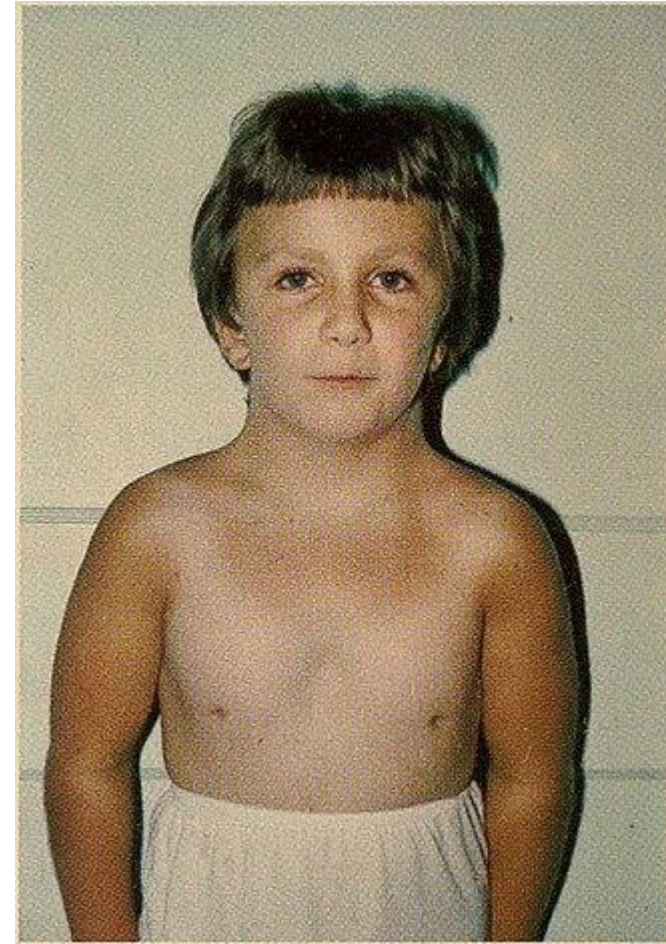
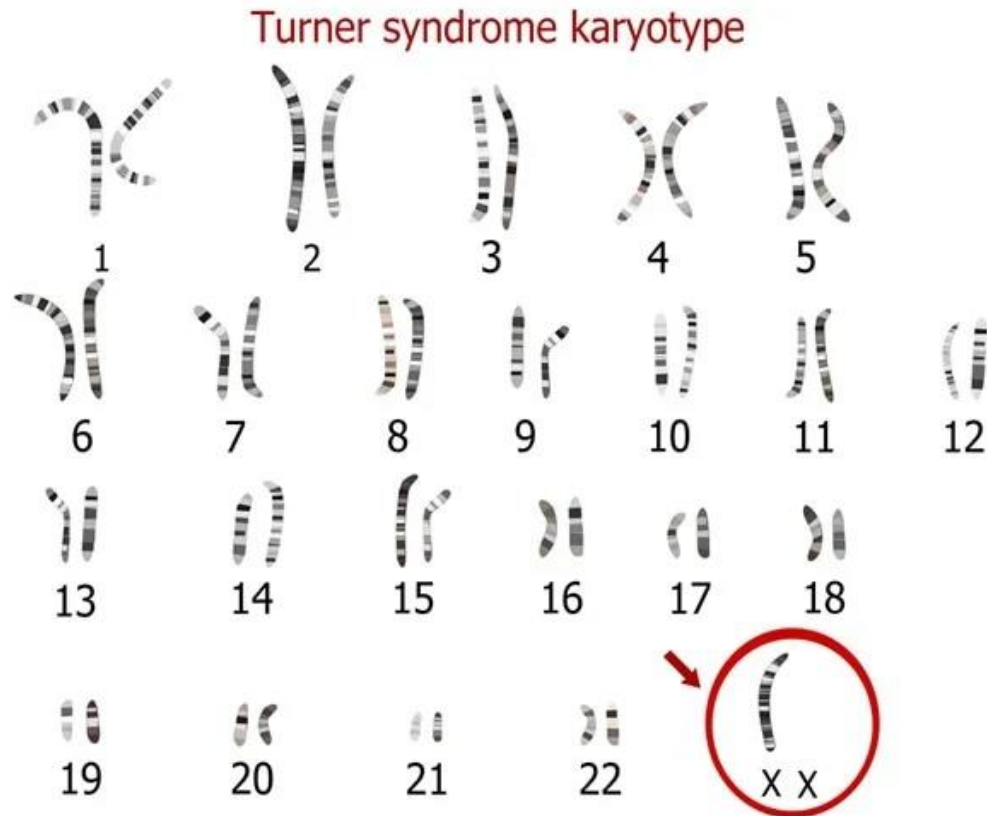


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Numerical Abnormalities (Aneuploidy)

2-Sex Chromosome Aneuploidy

- Turner syndrome (45,X)

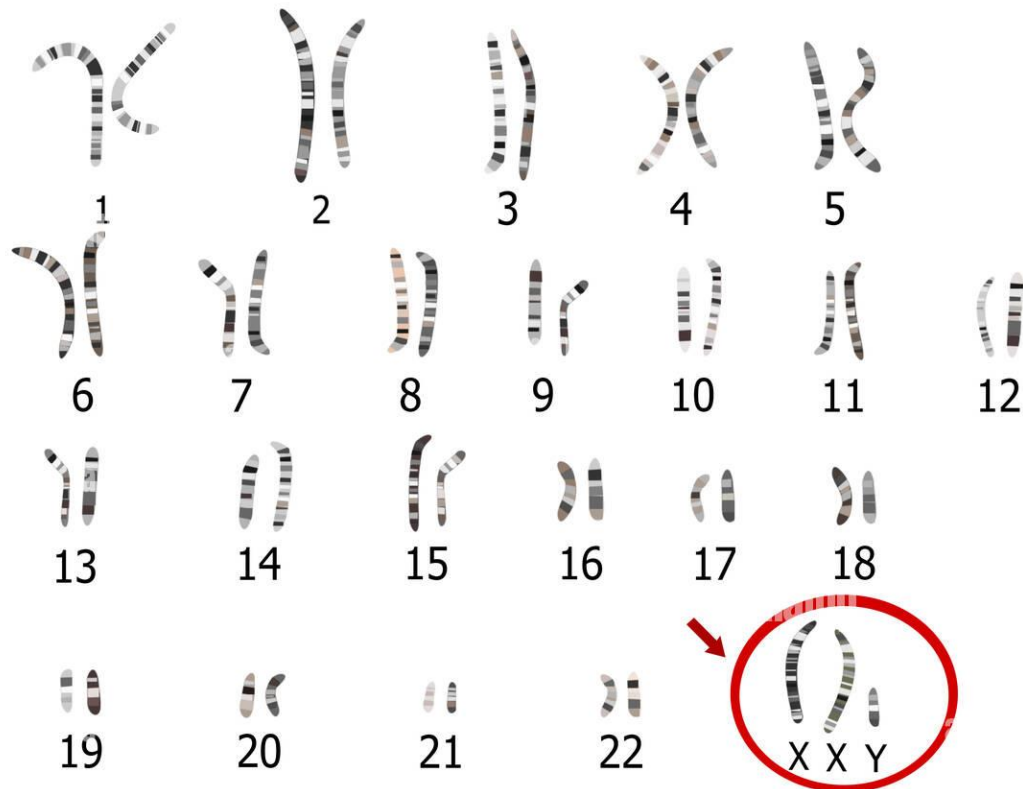


Numerical Abnormalities (Aneuploidy)

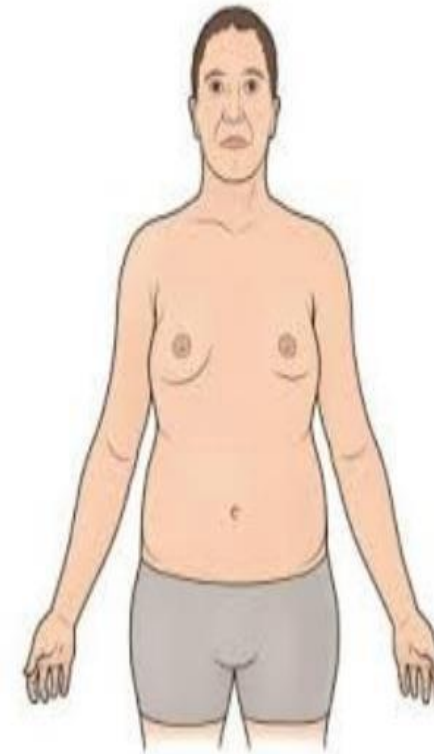
2-Sex Chromosome Aneuploidy

• Klinefelter syndrome (47,XXY)

Klinefelter syndrome karyotype



The signs.



- A taller less muscular body than males there age.
- Broader hips and longer legs.
- Larger breast.
- Weaker bones.
- A lower energy level.
- Smaller penis and testicles
- Delay in puberty or go a parcel amount.
- Less facial and body hair following puberty.

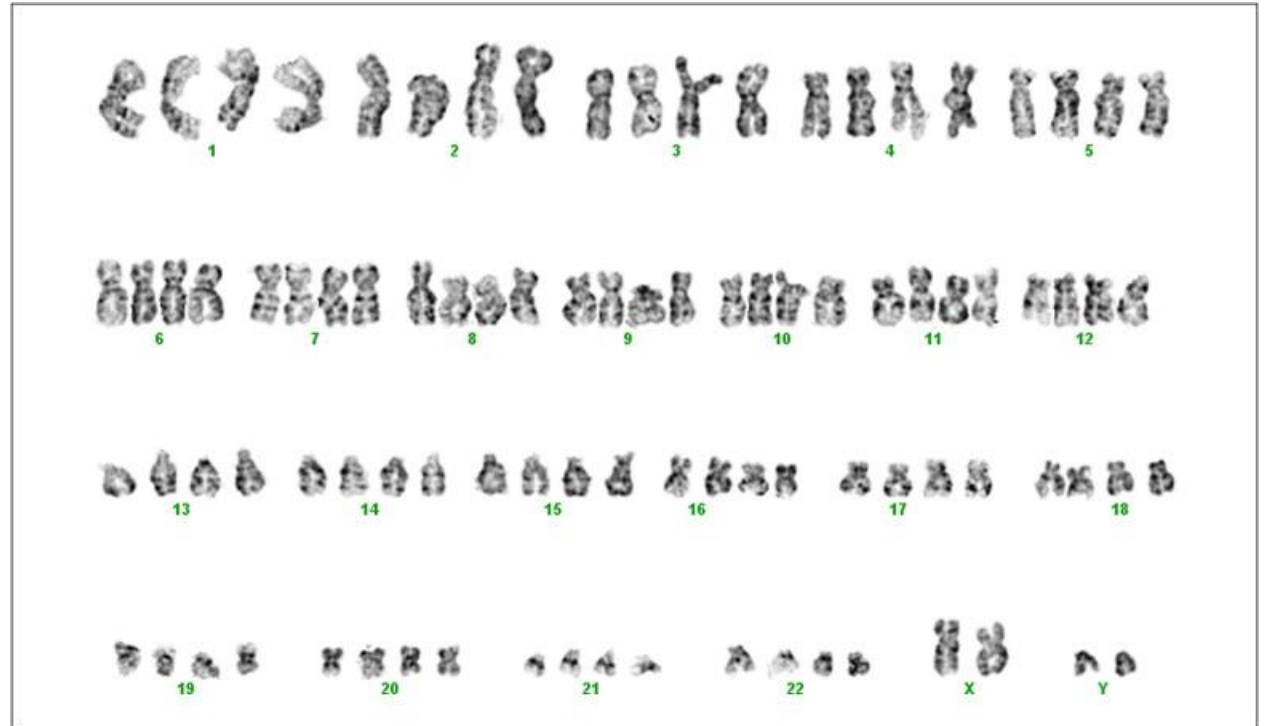
Numerical Abnormalities (Aneuploidy)

3. Polyploidy (rare, lethal)

- Triploidy (69 chromosomes)
- Tetraploidy (92 chromosomes)



Figure 15.35 A triploid karyotype. Courtesy of J.S. Haslam and K.P. O’Craft, Tameside General Hospital, Ashton under Lyne, UK.

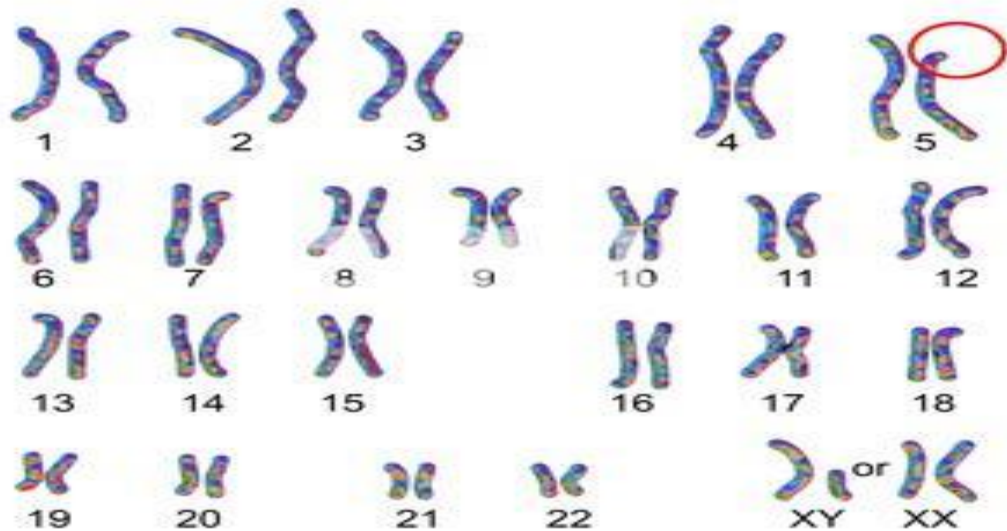


Structural Abnormalities

1-Deletions

Loss of a chromosomal segment.

Cri-du-chat (5p deletion)



alamy - 2CP4EAH

Structural Abnormalities

2-Duplications

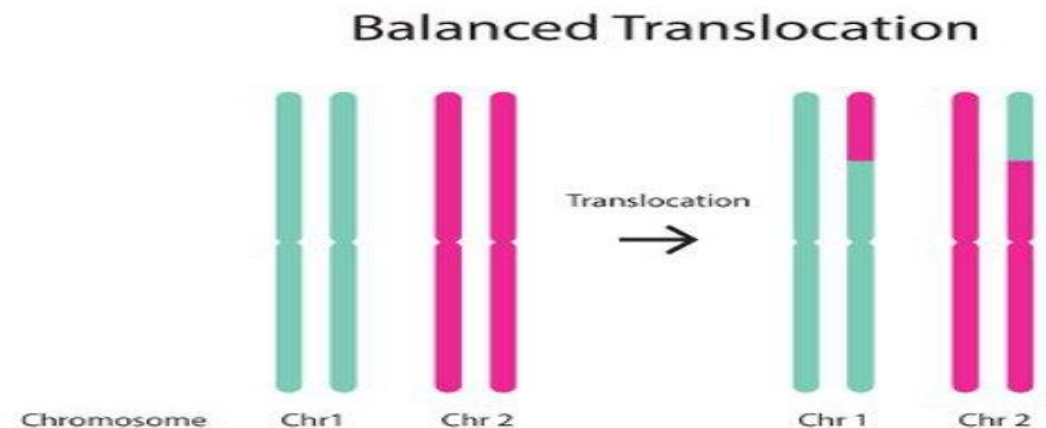
Extra copies of a chromosome segment.

3. Translocations

Exchange of segments between chromosomes.

A-Balanced Translocation

- No net gain/loss of DNA
- Carrier usually normal
- Risk of abnormal offspring



Structural Abnormalities

B-Unbalanced Translocation

- Gain/loss of genetic material
- Causes congenital anomalies
- Example: **Translocation Down syndrome**

Translocation Down Syndrome

Normal



14



21

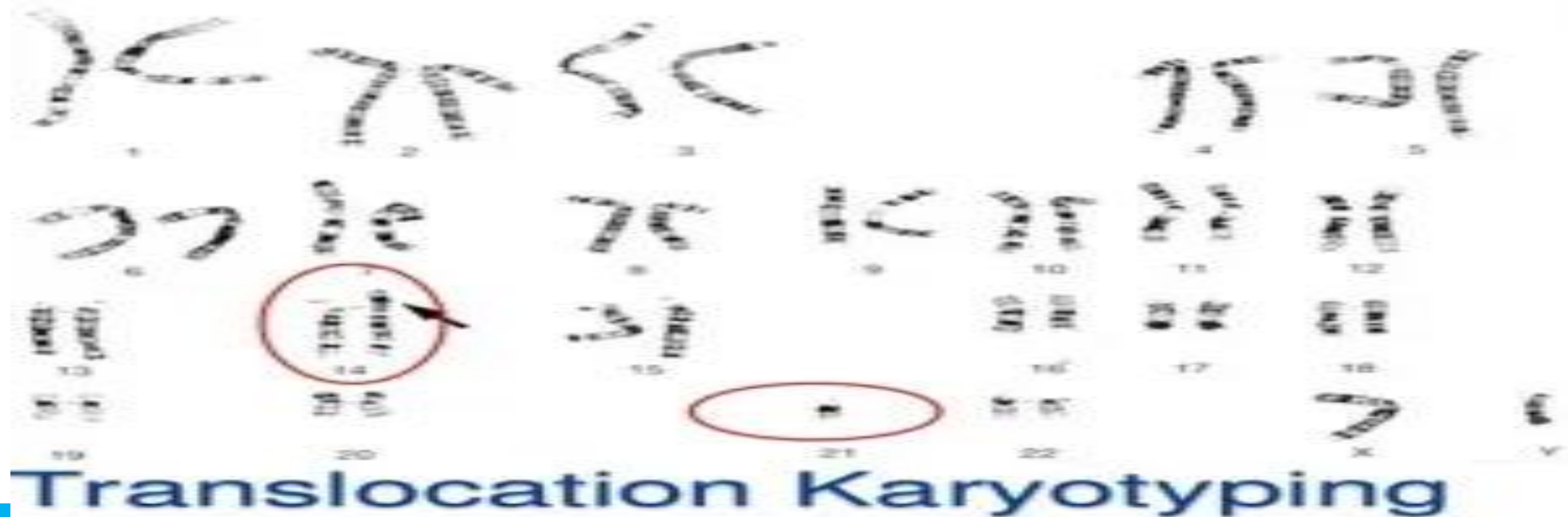
Abnormal



14



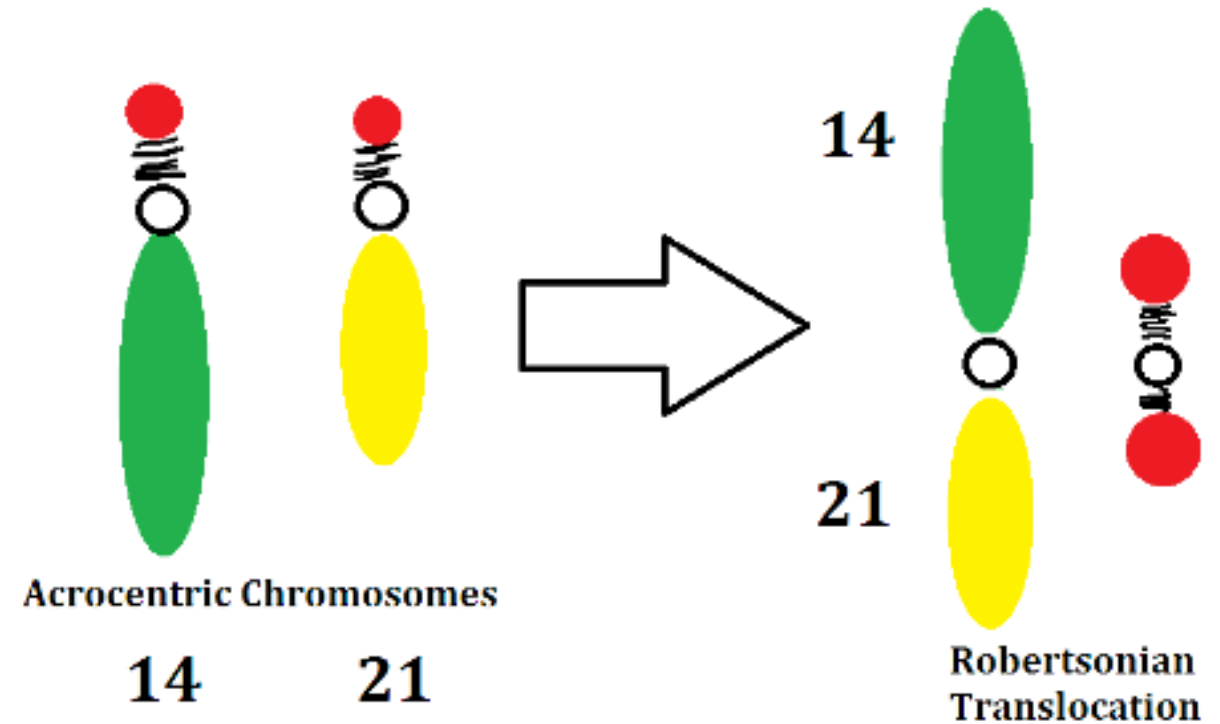
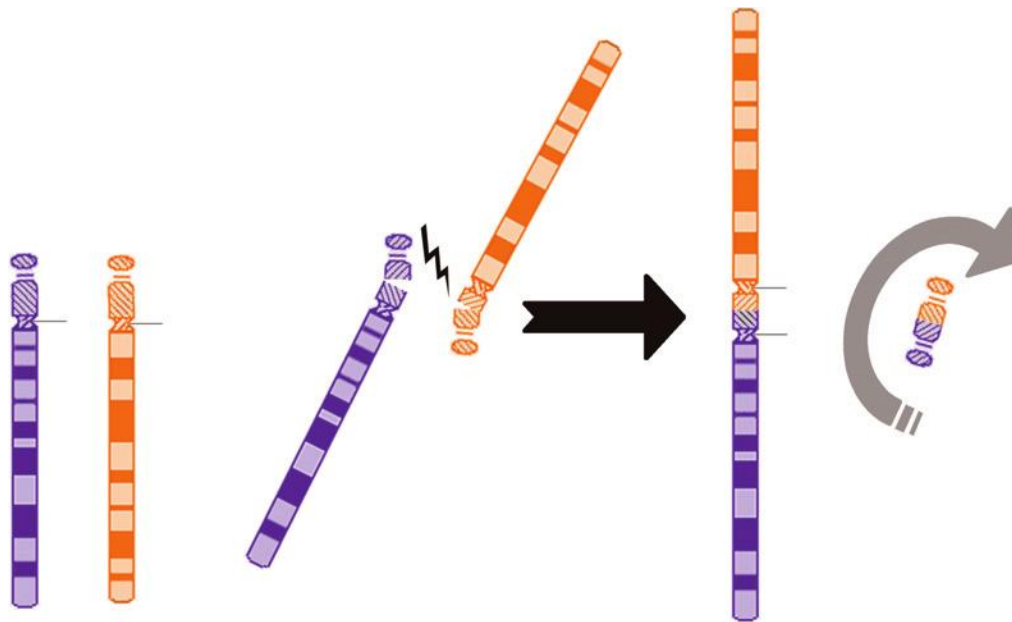
21



Structural Abnormalities

C-Robertsonian Translocation

Fusion of two acrocentric chromosomes (13, 14, 15, 21, 22).



Structural Abnormalities

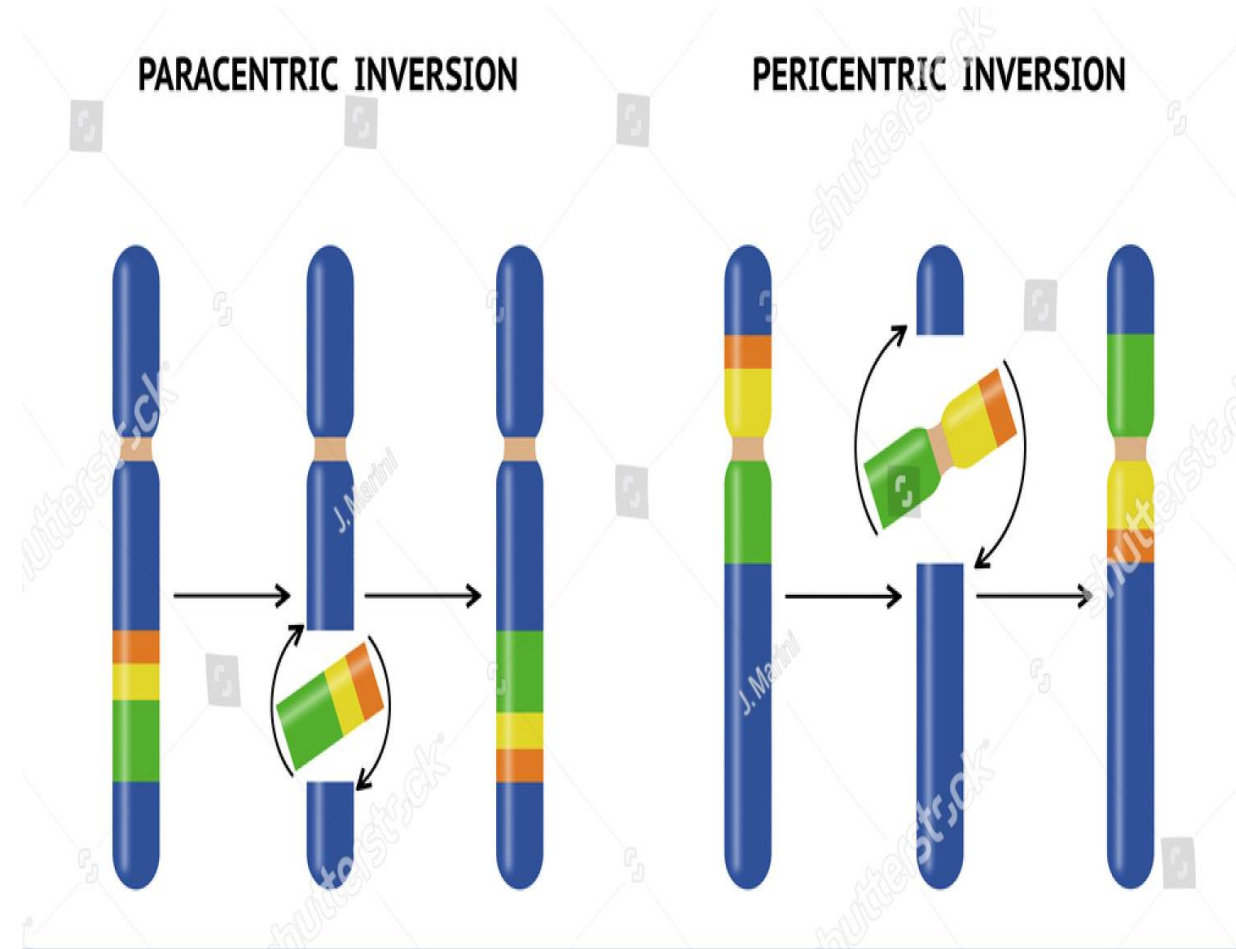
4. Inversions

A chromosome segment reverses orientation.

a. Paracentric (does not involve centromere)

b. Pericentric (involves centromere)

Often balanced, but can cause infertility or abnormal offspring.



Structural Abnormalities

5. Ring Chromosomes

Ends of a chromosome break and fuse into a ring.

Causes loss of terminal segments.

